



CORPORATE FINANCE

PRESENTED BY
Diva Agrawal

SOS 2025 / H07





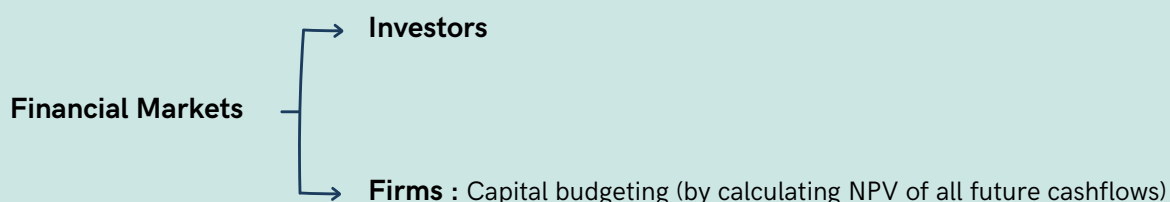
Table of Contents



<u>Perfect market</u>	03
<u>Rate of return</u>	03
<u>Stock and bond valuation</u>	06
<u>Time varying ROR</u>	09
<u>Yield curve</u>	11
<u>Uncertainty and Risk</u>	14



- **Time Value of Money(TMV):** Money today is worth more



- **Perfect Market:**
 - No taxes
 - No transaction cost
 - No difference in opinions
 - Many buyers and sellers
- Extra Assumptions:
 - Interest Rate
 - No inflation
 - No rise

The net return on a loan is called **interest**.

ROR on a loan is called the **interest rate**.

- Interest payments: has a maximum payment limit
- Non-interest payments: has unlimited potential.
Registration fee, late payment fee, penalties

Not every ROR is an interest payment(Eg, lottery).

BONDS	VS	SAVINGS ACCOUNT
You know the exact future value of money today.		The interest rates fluctuate, so you never know what might happen. These are like a sequence of one-day bonds.

Payoff of Return on investment is simply the amount returned(C_t)

$$\text{Net Return} = C_1 - C_0$$

Rate of Return(r) = $(C_1 - C_0)/C_1$. ie. Ratio of net return to investment.



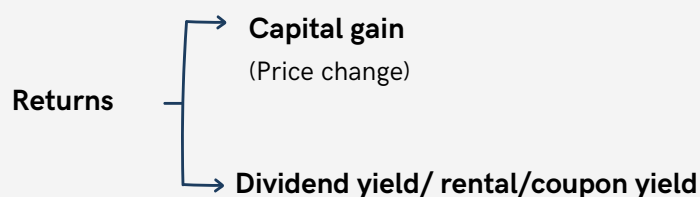
Interim payments:

Stocks - dividends for finding ROR, add interim payments to net returns

Bonds - Cash coupons

Real estate investments - rent

When there are interim payments:



- **Nominal vs Real interest rate**

(without
accounting for
inflation)

Basic points are 1% of a percentage.

For example, 10% → 10.5% → 50 BPS increase

10% → 15% → 500 BPS increase (or 5 points)

help with percentage ambiguities

Time value of money:

Say you earn interest on an investment, the same money today (A) will be worth more $A(1+r)$ in the future.

$$r = (FV - PV) / PV * 100 \%$$

TMV is not inflation, it is that fact that you can earn interest on money.

Compound interest is interest on interest

$$(1+r_t) = (1+r)^t$$
$$(1+r_1)(1+r_2) \dots (1+r_t) \rightarrow \text{spot rate}$$

Compound rates of return can be negative even if the average is positive, example, 200% and -100%

$$\text{Average} = 50\%, \text{Compound} = (1+2)(1-1) - 1 = -100\%$$

Effective Annual Rate (EAR) is the actual one that comes out after compounding.

APY (annual percentage yield) or AER (annual equivalent rate)



Annual Interest Rate = Daily Interest Rate * 365

Daily interest rate: It is the interest rate that gets compounded over the year to give the effective annual rate.

The annual percentage rate (APR) is the monthly interest rate.

Certificate of Deposit (CD) → long-term investment. Bank quotes APY.
You will earn more money when the APY is quoted.

Capital Budgeting: Allocating funds to different projects in a firm by assessing their present values.

$$r_1 = (C_1 - C_0)/C_0 \Rightarrow C_1 = C_0(1+r_1)$$

Discounting → reducing a future value to translate it to present value.
→ opposite of compounding.

Cost of Capital: The rate of return at which you can raise money elsewhere.

Opportunity cost: The best possible use of that money in the present economy.

Discount factor = PV/FV

- Price of bond and rate of interest (and ROR) move in opposite directions. ROR of bond decreases if price increases.

NPV is the present value of all the future cash flows minus the initial investment. If NPV is greater than zero, take the project. → NPV capital budgeting rule.

$$NPV = C_0 + C_1/(1+r) + C_2/(1+r)^2 + \dots$$

It is not necessarily better to invest in a fast-growing company; the stock prices already reflect their future cash flows.

Any sudden wealth gains accrue to past shareholders, not new investors.

Simple perpetuities are projects with constant streams of cash flows that repeat forever.

$$\text{Perpetuity PV} = C_1/r$$



Growing perpetuity is a perpetuity with cash flows, increasing at a constant rate(g) less than the cost of capital(r).

Applications:

- Cash flow is assumed to grow with inflation.
- In terminal values (pro-formas)

Applications → GORDON GROWTH MODEL

A company's dividend that it gives next year grows with a constant growth rate, and the rate of return is constant, then its stock price should be $D_1/(r-g)$ or less for it to be feasible to buy.

Dividend yield = $r - g = D_1/\text{stock price}$

→ **Plowback Ratios** = $(\text{Net income} - \text{Dividends})/\text{Net Income}$

ie, how much profit is retained rather than given out to investors.

It is often contemplated that the plowback ratio is proportional to the earnings growth rate.

Annuity: A stream of actual cash flows for a given number of periods.

$$PV = C_1(1 - (1/(r+1))^T)/r$$

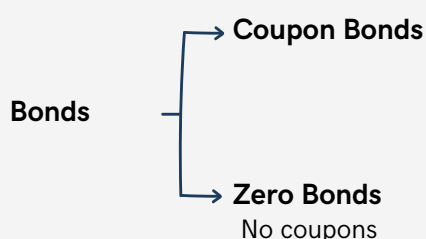
Application: fixed-rate mortgage loans.

Mortgage: A loan to buy real estate, where property serves as collateral.

→ They usually quote interest rate by dividing the annual rate by 12, not by compounding, that is, it quotes 7.5%/12 without compounding, but in the end, compounds 0.625 months, so it adds up to 7.7% per annum.

Two reasons to classify mortgage payments into principal and interest factors:

1. You need to know the principal due for an early payment.
2. Interest can be deducted from tax bills.



Level coupon bond: A coupon bond with constant coupon payments. They are named by adding up annual coupon payments and dividing them by the principal amount. For example, a '3% semiannual coupon bond' means if \$10,000 is the principal, then 150-150 will be paid every six months.

Bond trading at Discount → in loss

Bond trading at premium → in profit



QUOTES ≠ ROR

In bonds, the final principal is fixed while the bond price goes lower/higher.

The longer the bond, the higher the loss, and the higher the interest rate, the higher the loss.

Growing annuity: $PV = C_1 \{1 - [(1+g)/(1+r)]^T\} / (r-g)$

Free interpretation of money: If you buy something, it earns a profit and future, even after discounting, you will have earned money 'for free'.

Other info is redundant in a perfect market, but in real life, it is important to consider taxes, transaction costs, disagreement, shortage of options, and adjustment of the market.

In a perfect market, we can make investment decisions without the concern of consumption decisions. This is called separation of decisions.

Scenario/sensitivity analysis: Considering simple projects to learn how changes in estimates matter to the ultimate present value.

→ Analysis of short-term project:

10% error in cash flow → 10% error in PV

10% error in COC → Approx 1% error in PV

→ Analysis for long-term projects (let 30 year, 8% COC, 200\$):

Correct PV = 19.88

10% error in cash flow → \$21.86

10% error in COC → \$15.93

In PV estimation, cash flow is more important for short-term projects, while COC is more important for long-term projects.

Internal rate of return (IRR) finds the discount rate with the NPV set to zero. In bonds, IRR is called YTM (yield-to-maturity rate).

Yield-to-maturity rate of a bond is the same as its percent coupon bond.

Some projects have no IRRs:

→ with all negative or positive cash flows → No IRR

→ with some negative and some positive cash flows → Multiple IRR



IRR capital budgeting rule: If a project's IRR is above the discount rate, referred to as the hurdle rate in this sense, then the project should be taken.

NPV and IRR rule give the same results.

Why IRR over NPV?

- Can be calculated without knowing COC.
- Easier to estimate the 'buffer' for errors, ie, the difference between COC and IRR.
- Discussing the appropriate COC for finding NPV would cause issues.

Problems with IRR:

- Safer to use in PV, because it is easier to forget whether it is an investment or a loan. IRR is the same for a project with positive and negative NPV.
- Multiple or no IRR; if there are multiple IRRs, the correct one cannot be determined.
- No concept of scale. IRR se zyada profit matters.
- No obvious hurdle rate.
- Time factor: If a one-year project gives 5% return and 30 30-year long-term project gives 10% return, the short-term is better since it can be repeated.

Profitability index: PV of future cash flows/ project cost

Basically, one dollar will convert into how much.

- It also takes into account the initial investment.
- It is more useful for comparison, although it gets the same result as NPV, too.

Drawback:

- Favours low initial cost rather than viewing the actual profit (it should be used to compare projects with similar costs).

Payback rule: Projects are assumed to be better if money can be recovered faster.

Benefits:

- Simplicity
- helps set time boundation
- helps if you are an entrepreneur with limited capital
- usually correct
- accounts for time value of money

Earnings multiples: To compare the earnings of our project to those of other firms.

Accounting ROR: net income/book value of equity

Managers care more about earning (because it shows up in their evaluation, too).

**Time-varying ROR**

ROR is higher for long-term investments than short-term investments.

Re-investment rate: rate of return available after a project has been completed.

$$(1+r_{0,T}) = (1+r_{0,1})(1+r_{1,2})\dots(1+r_{T-1,T})$$

T years. Spot rate. Forward rates

holding ROR

Annualised ROR:

$$(1+r_{0,T}) = (1+r_T)^T$$

A convenient rate, which measures the rate at which money accumulates. This makes it easier to compare projects.

Averaging vs Annualising:

Value	\$100	→	\$200	→	\$100
	T_0		T_1		T_2
	100%		-50%		

Annualised ROR = 0%

Average is deceptive, often shows

Average ROR = 25%

more, simply because AM>GM.

→ Often, average is quoted instead of annualised.

Maturity: Date of very last payment.

Duration: Payment weighted average of payout times.

$$\text{Duration} = C_1 \cdot t_1 / \Sigma C + C_2 \cdot t_2 / \Sigma C + \dots$$

Macauley Duration: Replace raw cash flows with their discounted prices.

These parameters try to show when the average payment will occur.

Inflation: the process by which prices of goods increase over time, or money loses its value.

Nominal terms mean simply ignoring inflation.

Real terms (Inflation index terms are those in which inflation has been taken into account)

To measure the inflation rate(π) in the USA, the BLS has invented bundles/baskets of household commodities called CPI (consumer price index).

Inflation rate mirrors reality. In the US, lower the inflation, higher the government has to pay to social security recipients. Yet, inflation rates are often overstated. Why?



→ Technological advancement: Prices remain the same while quality improves => value of money has decreased.

→ Retail stores offer products at a cheaper price, so commodities are actually cheaper than their MRP.

The CPI just checks price and not quality.

PPI is the production price index. It takes into calculation the wholesale price of the said 'bundles'.

$$1+r_{\text{real}} = (1+r_{\text{nominal}})/(1+\pi)$$

Real interest rates are often negative.

In NVPs, it is advisable to use cash flows adjusted according to inflation; that way, there won't be any confusion, and we can use the nominal discount rate and its calculation.

Gold → not a great measure because of fluctuating prices, but good for historical comparison.

Treasuries: A marketplace for bonds issued by the government to cover its national debt. No risk since the government can just print more money.

A bipartisan problem - National debt: because the governing party spends but doesn't expect that it'll have to pay the debt back themselves.

In late 2021, the US government owed the US Treasury obligations more money than the whole US GDP.

Ownership of the Treasury bonds: 50% US investors
50% foreign investors

Interest rates may vary every moment, so the rate of return of bond prices can never be predicted. So, treasuries guarantee a fixed income, but not a fixed ROR.

**Yield curve/term structure:**

It is a graphical representation with time to maturity on the X-axis and annualized interest rate on the Y-axis. It shows whether long term investment have a higher annualized ROR or short term. The graph is usually upward sloping, because long term investments offer higher interest rates.

Types of yield curves:

Flat - Shows that there is no difference between short term investments and long term investments. Here, to calculate rate of return we can use the formula for compound interest:

$$(1 + r_{0,t}) = (1 + r_{0,1})^t$$

Upward sloping - Shows that long term investments offer a better return than short term investments. The curve is steeper when the economy is emerging from recession.

Downward sloping/inverted - Shorter term investments are better than long term ones. This shows that the economy is impending towards recession.

Humped - ideal investments are those which are neither too short nor too long.

Corporate projects have to make yield curves too, using the Treasury yield curve as a benchmark. It is so because they have to compete with the government for funding.

In a perfect market, with no transaction costs or penalties, there is no link between when we need the money and the maturity, because the bond can always be sold before its maturity.

How time to maturity affects risk:

A company is much more likely to go bankrupt in the next 10 years, than the next year. This is why short term bonds are always safer. This is not the case for Treasuries, because there is no risk in repayment.

All paper losses are real losses, except the paperwork we do for taxation purposes.

Term premium/time premium is the difference between the long term and the short term annualized rate of return.



Statistics

Expected value (ex-ante) - It is the weighted average of all future outcomes.

Realisations (ex-post) - The actual outcomes.

$$\begin{aligned}\text{expected value} &= \sum(\text{probability of an outcome} \times \text{value of the outcome}) \\ &= 3.5\end{aligned}$$

Realisation could be any integer between 1 and 6.

Random variable is a variable whose outcome is not yet decided.

Fair bet - A fair bet is one whose cost is equal to its expected value.

Reward - A measure of what we expect to get from our investment (Expected value).

Risk - A measure of how spread out our possible outcomes are (mean deviation).

Risk-neutral investors are willing to take risks.

Risk-averse - when investors expect some compensation for the risk involved.

We take less risk and are more averse to smaller bets; this is why in financial markets, there is 'risk sharing', that is, investments are divided into shares. Therefore, the aggregate market is less risk-averse than individual investors.

Most loans are not risk-free, except the treasuries (in the US)[ie. government bonds]

In loans, a solvent is someone who will be able to pay back the loan; a defaulter is someone who will fail to do so(bankruptcy).

Treasuries offer 10% and money lending offers 45%, then

Time premium - what treasuries pay is 10%, while the remaining 35% that you expect just to be even, is called the default premium (what if the person goes bankrupt?)

The quoted rates and YTM are always the promised interest rate, not the expected rate.

Promised interest rate = Term premium + Default premium

Usually, the default premium is zero.

Risk premiums: The interest you gain for the risk you are taking.



Imperfect market premiums: Pay you for future difficulties in selling your bonds.

Credit rating: A rating to make it easier for lenders to judge the probability of default.

The top rating grade is known as the investment grade, while speculative grade is called junk grade.

Ratings have limited usefulness:

- Common risk is not considered in the comparison. The investors have to face much higher losses when there is a common risk than if one of their bonds went default.
- Rating agencies are not liable to their ratings, so the ratings can be deceptive.
- They are paid for their ratings so naturally, ratings can be bought.

Riskier bonds have higher promised ROR.

Despite promised rates varying so much, the *expected* ROR is quite similar to the promised rate of return of the treasuries.

Interest rates vary with location, LIBOR (London Interbank Offered Rate) is used as a benchmark. The market is gradually shifting to SOFR (Secured Overnight Financing Rate).

Credit Default Swaps (CDS) essentially loan insurances. If the borrowing company goes bankrupt, the lending company does not have to suffer losses, it is covered by the insurance company in return for insurance premium.

The third party may also place a bet on the borrower not going bankrupt, so it does not directly lend money but still has a stake in the transaction.

But if a company loses money, its credit ranking goes down because its stored capital gets depleted, making it riskier.

The investment firms may pressurise the borrowing company to declare bankruptcy to receive the insurance money.

The credit rankings can be bought.

The issue with CDs is that if the market crashes, taxpayers will suffer.

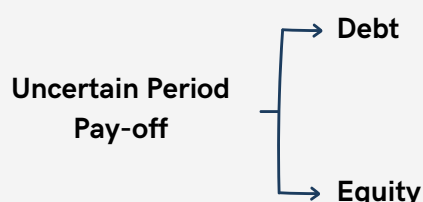


Uncertainty in capital budgeting: We have to compute present values with imperfect knowledge about future outcomes.

Pay-off table (State table) assigns a probability to the project value in each possible future value, relevant scenarios.

Loan/Debt/Leverage is the amount to be paid back.

Levered ownership/Levered equity/Equity/Stock is what accrues to the business after paying out the debt.

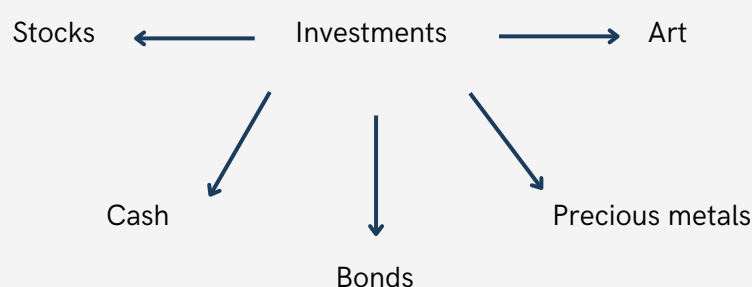


Debt has to be paid off first; it is the safer option and has the right to ownership of the project if it goes into default.

Equity reaps a higher ROR but has a higher risk. But equity holders have residual ownership.

No-recourse loan: If the project goes into default, the creditor can only seize that project, but not any of the borrower's other assets. This is called **limited liability**.

Investments:



Indexes: Describe the value evolution of stocks based on their portfolios and give investors basic guidance. For example, the S&P 500 in the US, the Nifty 50 in India.

S&P 500 → 500 largest publicly traded firms, value weighted in one portfolio.

Market capitalisation – the number of all outstanding shares times the price per share.



Complication: These indexes ignore dividend payouts in calculating ROR, so the ROR is basically perceived as a benchmark. But firms may try to maintain a portfolio similar to those companies listed and not give dividend payouts.

The difference between the average and analysed compound rate rises due to volatility in the market. Quoting average ROR makes the risk project seem better.

We should always make simultaneous investments over sequential ones (Arithmetic vs Geometric average)

For a bell-shaped curve, then: $AM - GM = Var/2$

Value stocks → Older firms with higher book values than market values.

Growth stocks → Younger firms with the opposite pattern.

Earlier, value investing was prominent, with growth stocks being attractive to investors due to high promised RORs, allowing the value stock to perform better. This was called 'value investing'.



Beta is a statistical measure that quantifies the volatility of an individual stock or portfolio with respect to broader market

Stock Infotech financials and materials give good or while energy and utilities paid off polarly in the last 17 years wine and art even resulted in losses.

The figures for art and wine are given out by organisations publicising their interest, so the actual are ROR may be worse.

Venture capital fund startup startups need venture capitalist because they may not have access to capital markets or bank loans the investors may get a voice and company decisions there get exceptional rate of returns but have a higher risk.

Private equity is offered to companies which are well established but do not trade publicly.

BCS and p is I high risk, hi free investment vehicles with long capital lockup periods. Some issues they might face are:

Mark to market prices are guess work
Investments are not easily transferable
Past holding our war is difficult to come by.

Comovement: looking at performance of other assets classes in comparison to the stock market.

Return series: A plot with rate of return of stock index on X-axis while ROR of second asset on Y-axis. Points name relevant years. The 'linear regression line' is the best fit of all these points and its slope is called beta.

Comovement is important. Investors take up an investment even if the ROR is lower if it moves opposite to the market, so that they have assets when the market crashes.

Beta is a good measure for co movement:

1. $\text{Beta} > 1$ (slope was steeper than stock market): when stock market did good, it did a lot better but when stock market crashes, it would drop even more.
 2. Beta lies between 0 and 1 : Scales down the blow when the stock market drops.
 3. $\text{Beta} < 0$: games value when the stock market drops, which means it acts like insurance.
- Beta of market is called market beta and it is always equal to one.



Correlation has a limited range of -1 to 1.

If correlation equals to 1, they move in opposite directions

If correlation equals to zero they move independently

If correlation equals one they move in perfect sync

Advantage: it is easier to judge than beta.

Disadvantage: no concept of scale even if there an asset moves very slowly with the x-axis the correlation would be 100%.

Finance and history: If an asset has great returns in past it could be a great investment or just over price due to a bubble. Bubble is a sudden rise of an asset price which leads to crash (or burst) in the future.

Tail event : An event that seems completely inevitable.

Factors are broad persistent drivers of return. Factor investing: keeping in mind factors, choosing passive management (buying and holding larger portfolios or asset classes) over active management trying to project individual stock.

ROR is difficult to product but risk is easier still standard deviation that quite stable.

Estimation tips: SDs are better predicted from higher - frequency data.

Whatever the prediction is, it is better to average it out with the market beta.

Correlation and Causativity: Just because two variables are related to each other, does not mean one was caused due to the other, this would be a spurious correlation. There might be a third factor causing both of them to change similarly.



Brokers have mainly two tasks:

1. Traders - Handle trades.
2. Prime brokers - Keep track of holdings, buy on margin, short sell assets.

Market has three parties:

1. Investor - Gives money
2. Broker - places order
3. Wholesaler: Buys shares, and if they're unavailable, holds the money

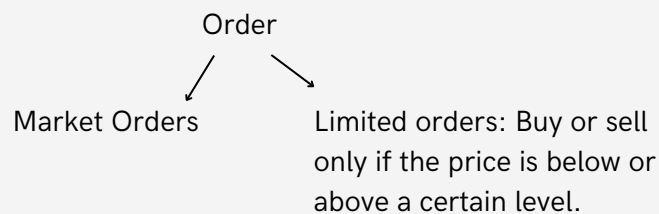
Auction markets are less active venues for buying and selling. They have lower execution cost but slow execution.

Execution costs:

The payment for order flow is an arrangement in which the broker earns the difference between buying and selling price also called the bid-ask spread.

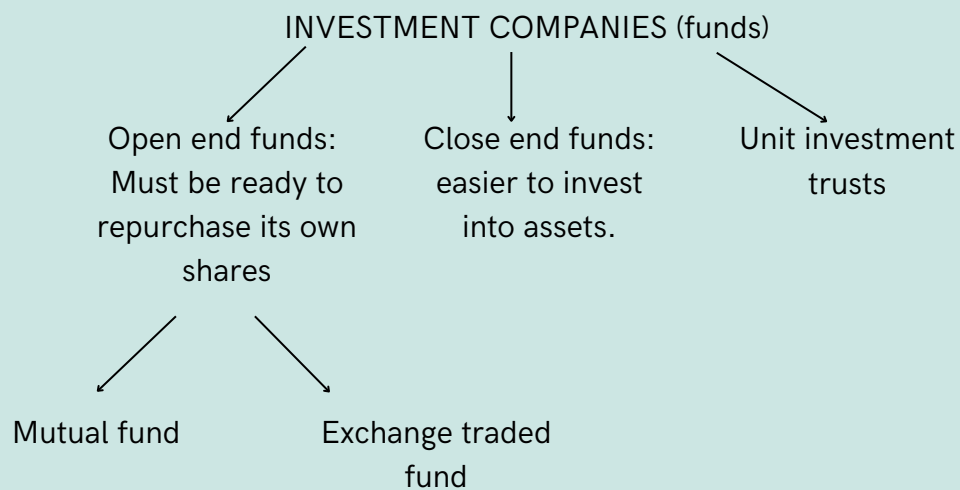
Day trading when did buy and sell stock the same day to avoid having lost money the next day due to changing interest rate

High frequency traders profit through higher trading volumes



HFTs buy limit orders very quickly, before the trader can revise it. Nowadays HFTs have to outrun their competitors which has eroded their profits.

Sorting stocks: Selling at higher price and buying at lower prices.



A mutual fund must post its net asset value at the end of each trading day.

Exchange traded funds have rules about who can ask for creation or removal of shares it traits like a normal company on stock exchanges.

American depository received (ADR) is a passive investment vehicle that owns a foreign security stock, held escrow at US bank, making it easier to trade and foreign security without incurring losses due to transaction costs.



Portfolio risk describes the potential loss of value.

Portfolio diversification is based on imperfect correlation or non-synchronicity among investments. It helps reduce overall portfolio risk.

If two random draws are uncorrelated then the sum of these two draws has a variance equal to the sum of their variances.

This means that risk increases slowly as compared to rate of return.

For example a stock has a rate of return 10% and risk 20% if we want to take it up to 2 periods:

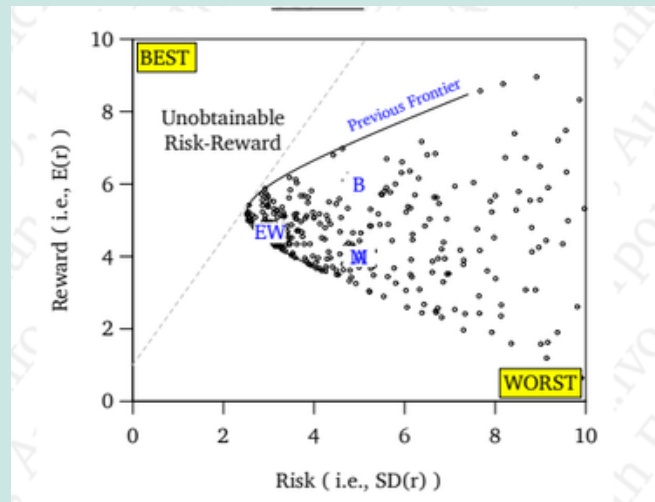
$$\begin{aligned} \text{ror} &= 20\% \\ \text{Variance} &= 20 \times 20 + 20 \times 20 \\ &= 800\% \\ \text{Standard deviation} &= 28.28\% \end{aligned}$$

Here, mean r o r double but SD only increased 1.41 Times

This implies risk grows approximately with a factor of root square of time.

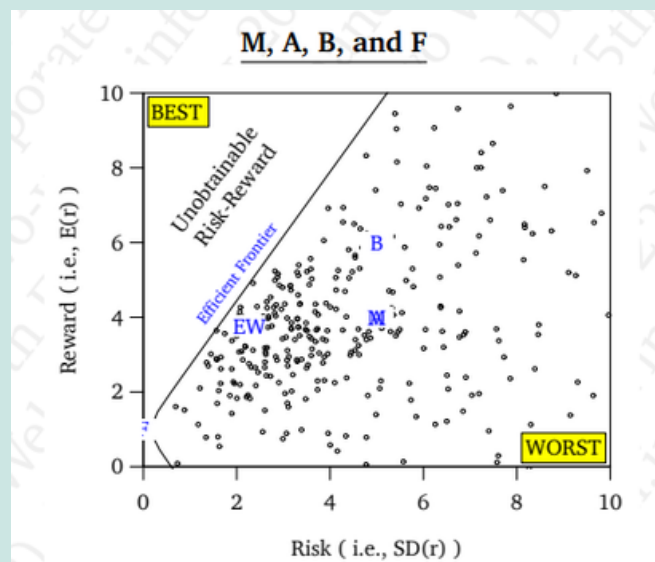
Sharpe ratio is a ratio of reward to risk ignoring the compounding, sharpe ratio is $(e\sqrt{t})/SD$

Mean variance graph is plot with expected rate of return on y axis and risk on x axis each point represents a portfolio more the portfolio northwest the better.



It is always a hyperbola for multiple assets. The most northwest boundary is called the effective Frontier. The lowest risk portfolio is called minimum variance portfolio. All the portfolios lying on the effective Frontier are better than those lying in it. The area outside the frontier is called the unobtainable risk-reward.

If we add a risk free asset our plot will look like this:



So to find the perfect portfolio, we use an intersection of these two graphs.

Drawing a tangent to the effective frontier from the rate of risk free return we get our ideal portfolio also called the tangency portfolio.



Smart investors have portfolios nearing the market; therefore, it is beneficial to consider the beta, how the ROR of our project correlates with the market.

Alpha: it is the intercept on the return series shows the return on top of the market return, it is an indicator of whether the risk is worth taking or not.

Alphas of only similar companies should be compared since they also depend on the benchmark. Alphas are very sensitive to change.

Covariance is the probability weighted average of the products of two net of mean variables.

Beta of C with respect to beta of M is the ratio of covariance of M and C, and the variance of M.

This method of finding beta is called linear regression

Real world market beta estimation

- It is better to use daily rate of returns.

- 2 to 3 years of daily historical returns should be used; anything older than 5 years is irrelevant

- It is better to shrink the estimated return towards the market beta by averaging

While working with historical data to account for the irrelevance, instead of dividing by n , we divided by $n - 1$.

A portfolio that invests equally in all its assets is called an equal-weighted portfolio.

Value portfolio is one in which weights correspond to the market values of its components.

We can compute value-weighted average to find expected rate of return and beta, but not variances or standard deviation.

Benchmarking cost of capital

Investor preferences:

- Long-term investments are undesirable

- Equity is undesirable over bonds

Corporations generally use the risk-free rate or Bond rates with similar durations for discounting their cash flows, although it is better to use similar companies to estimate the discounted rate.

Projects can be seen as a combination of safe (debt claimed financing) or risky (equity claimed financing).

Equity premium is expected or over what risky equity projects offer above what risk 3 points are equity market premium is equal



Methods for equity, premium estimation:

1) Historical averages I: equity premium is the average of past premiums. Stock returns have two components:

Term premium

Residual pure risk premium

Equity premium = Form premium + pure risk premium

Equity premium puzzle: the rate of return offered by the stock market is surprisingly high, and comparison to the risk it holds.

Some points to consider are:

Arithmetic means are often misleading, because they ignore the effect of volatility on compounding.

This could also be a term premium puzzle; long-term bonds offer a much higher ROR than short-term ones.

Sample period: although a larger set of data provides more accuracy, market returns change with time; old data can become irrelevant. Also, it is not necessary. USA will perform that exceptionally, so for estimations, it would be better to use a country with normal performance.

2) Historical averages II:

High rates of returns will attract more investors to stocks, and this will reduce their price, leading to a loss. In extreme cases, the turns into a bubble-burst situation.

3) Current predictive ratios:

Estimating a statistical regression to predict next year's equity premium/dividend yield.

This method does not work as well.

4) Philosophy:

How much return would be enough for investors to choose stocks over bonds? How do we balance the risk and reward? These questions lead to an estimated geometric average of 1 to 3%, which is quite reasonable.

5) Surveys:

Collecting estimates given by professionals to find the average expected rates of returns. These estimates are also made according to the previously mentioned methods and what financial books prescribe. The estimates are 3 to 4%.



6) Internal cost of capital

Internal cost of capital is the COC that company analyst use to satisfy their investors. The ICC must be equal to stock returns. So an average of these ICC estimates throughout the business cycle is taken.

Equity premium, short-term estimate can be used to predict other premiums.

Decomposition of stock market return into term premium and equity premium: if fractional part of stock is x , then the return is:

$$TP + x(MRT - TP)$$

Asset versus equity cost of capital:

Debt is always safer than the project, and equity is always risky.

Even if equity returns are high, the expected overall return may not be as high, because some part of the holding would be debt.

Asset pricing model: comparing our own opportunities to the price at which similar ones are available elsewhere.

CAPM:

Assumptions:

1. Investors are risk-averse.
2. Market is perfect.

Capital asset pricing model gives us a formula that relates an appropriate expected ROR for our project by feeding in the market beta, risk-free ROR, and expected return on stock market

CAPM ignores the standard deviation because smart investors can simply use it to diversify their portfolios; they care more about comovement.

CAPM limits are attention to just two benchmark assets. It gives us a major idea of where our project lies on a spectrum between stocks and bonds.

Economic equilibrium model is a justification for CAPM. In this view, financial markets are perfect, every investor faces, some tradeoffs, and uses the model, every asset is priced according to it.



CAPMS graph as the security market line (SML), with expected ROR of project on the Y axis and market beta on X axis. Its slope is the equity premium.

Issues with CAPM: the equity premium and beta are both estimated; therefore, it is hard to find out the actual values.

Debt beta is not necessarily zero. Companies and financial distress have unusually high leverage.

Conglomerates: firms with widely diversified and unrelated holdings. Diversification does not add value because investors can simply diversify their own properties.

Risk-free rate is also different for long-term and short-term projects, so we can create multiple SMLs.

Market beta is usually negative for long-term investments, so CAPM proves to be unreliable. Long-term investors refer to the curve instead.

Betas themselves are reliable only for short periods.

For estimating betas from historical returns, we can run a market model regression.

For a good estimate:

1. Daily returns are taken.
2. 1-5 years old data is taken.
3. A shrinking of 30% for short-term and 40% for long-term projects should be done towards the market.

Beta can be used for sorting stocks to estimate risk. Example: if we have invested amount C in an asset with beta B, we short an amount of BXC from the market to neutralise risk. This is called inoculating or hedging a project to the overall stock market.

Two factors that should be considered while shorting are:

1. Estimation errors in beta.
2. Idiosyncratic risk tolerance.

Equity risk premium matters only when a company has not immunised itself against market risk.

Shorting is like going against the stock market; it is generally not advisable.



Most investors do not hold diversified stock market portfolios; here, the CAPM collapses.

- House owners should find stocks that reduce house risk exposure.
- One should hedge one's career; the stocks go up when the value of one's expertise goes down.
- Entrepreneurs have a much higher COC, because they are always short of funds.

Beta should only be used for reducing risk, not for estimating returns. In the CAPM formula, beta should be kept 1.



Imperfect Markets:

One assumption that falls is that borrowing and lending rates are equal. Now, the rates of returns have a large range, not a fixed value.

Assets are worth more to seller than buyer. Here is an example:

Let's take a share of \$20, and two years later, its value becomes \$80. Tax rate is 20% and discount rate is 5% then tax for 2 years later is equal to \$10.88. If someone buys the stock midway when the price is \$50, tax would have been \$11.16. Therefore, the buyer loses more in tax.

Perfect market assumptions and their violations:

No difference in opinion: borrowing rate is higher for those with more chance of default.

Infinitely many investors and firms: if a single bank is dominant, it may keep a large gap between borrowing and lending rates, but since there is no other option, we will have to accept unfair deals.

No transaction costs:

To process paperwork, transaction costs are to be paid. Another cost we have to pay is our time in making deals, finding investors.

No taxes:

Savings rate is lower than borrowing rate because the interest earned is included in taxes, while no taxes are deducted on interest paid.

When shares are traded, the tax paid is higher, leading to an increase in the value of those shares.

Documents need to be filled out for government regulation of transactions.

In an imperfect market, separation of ownership and value breaks down. The project value is not unique anymore; it may depend on the owner.

The personal value for an item is what we are willing to pay for it. The difference between personal value and market value is called surplus. Marginal buyers are those who have no surplus.

Uncertainty can occur in perfect markets too. There can be different lending and borrowing rates due to the probability of default, but the expected return must be the same.



There are three mechanisms to alleviate information differences

- Covenants: contractual agreements that specify upfront what the debtor must do to maintain credit.
- Collateral: assets that a creditor can repossess if the payments are not made
- Credit ratings also help in the assessment, but they are given based on past repayments, not based on future estimations.

Behavioural finance: all investors and executives are humans with biases; the most important bias is overconfidence.

Transaction costs:

Real estate:

Direct cost-brokerage commission, advertising, insurance, house, inspection, local land registry.

Indirect-opportunity cost(could have earned money by renting it out), time cost.

Financial securities:

Direct cost-payment for order flow, bid-ask spread.